

Comparison between Touchscreen Technologies

Type of touch technology	Pros	Cons
Resistive touch	Touch can be sensed in any form, resistant to water and dust, inexpensive	Easily scratched, no multi-touch, passes less display light, less durable
Capacitive touch	Multi-touch, resistant to water, dust, moisture, scratch, very long life, good-quality display	More expensive than resistive, can detect touch only by objects that emit electric flow
Infrared touch	Lets through 100% light from display, most durable, digital sensor output, highly sensitive	Very high cost, single touch

durable than their counterparts.

"Capacitive-touchscreen phones undoubtedly offer the best quality. They are extremely sensitive as compared to resistive-touchscreen phones. However, with the evolution of technologies, even resistive touch displays have become smoother," says Kaushik.

Further distinguishing the resistive and capacitive technology, Kaushik adds, "There is quite a bit of delta in the cost in case of capacitive touch display. You pay \$3-\$4 more per square inch (46 to 62 cents more per sq cm) as compared to resistive."

The variance-projected capacitive touchscreen has been adopted in various high-end smart phones and handheld devices. "They are highly sensitive to touch and accurate, besides having long life even after rough usage," shares John Wu, marketing head, Fortrend. The capacitive touchscreens also offer certain advantages like anti-scratch, water-proof and high transmission rate.

Know your choices well

You may have heard terms like full touchscreen, touch plus keypad, touch with slider QWERTY, stylus options, etc, with respect to the touchscreen technology. But before you decide to buy a touchscreen, it is important to understand your requirement, how much can you afford, and which of these types are convenient to use.

A full touchscreen is the one in which the entire front panel display is touch sensitive. There is no keypad on the top panel. Whereas, a display that is in the form of a bar, and has a regular keypad, is referred as touch

plus keypad.

Phones like Palm Pre and Motorola Droid, come with built-in QWERTY slider. It offers you the advantage of both full touch display and keypad, but at the cost of more bulk and weight.

"Ideally, 2.4 inch (6.1 cm) is suitable for a touch keypad display, while a minimum of 2.8 inch (7.1 cm) display is suitable for full touch display," explains Kaushik. "If you want to type SMSes, you need the letters to be big in size. In this case, 3.2 inch (8.1 cm) touchscreen is suitable for you," says Rahul Sharma, CEO, Micromax.

According to Kaushik, the most suitable form factor is a bar type phone. Agreeing with him, Rahul says, "A slider adds to the volume and weight of the phone, and makes it difficult to carry."

Talking about display, Kaushik says, "Pixels are very important for the display as they decide its quality. The display quality and the response from the software is based on the resolutions and the pixel size." There are a variety of displays available in the market like QVGA, WVGA, HVGA, etc. A minimum of QVGA is required to have a clear and crisp display.

User interface or the software that supports the interaction also play important roles in touchscreen phones. They

Manufacturers of touchscreen phones

Acer, Apple, Asus, Blackberry, Fly mobiles, Gee Pee, HTC, i-mate, LG, Micromax, Nokia, Samsung, Palm, Sony Ericsson, Spice Mobile

decide the response time of a phone while processing functions. The touchscreen functionality is driven by software that interacts with the user. There are certain semi-touch media phones where touch is enabled for certain applications. "The functionality is driven more by consumer insight. In terms of touch, consumers would prefer using it for media functions for the ease of operation. We have enabled this in our semi-touch Aino," shares Dhiraj Soni, marketing manager, Sony Ericsson.

One has to understand that all the technologies have their own power needs. Most of the touchscreens may not provide extremely high battery life. This does not mean that they are vanilla. While selecting a touchscreen, you should look for a model that offers at least 3.5 hrs of talktime. The standby time does not really make much difference.

Summing up, Arora says, "To enjoy the touchscreen feature, the phone should have a big screen, high resolution, interactive user interface and above all high-end processor back up."

Regarding durability and reliability, Kaushik says, "The only way to fix your broken touchscreen is by replacing it." So, do not forget to buy a case to protect your touchscreen phone.

Future touchscreen technology

There is another touchscreen technology which is not very popular in the mobile phone segment. It is based on infrared technology and is the most sensitive to use. It emits 100 per cent light, delivering high-quality image. Another significant feature is the digital form of its sensor output, which eliminates the complex signal conversion and processing required in other technologies. Its only drawback is that so far it does not support multi-touch. Also, it is most expensive as compared to other existing technologies. But experts see a great potential in this technology. ●

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